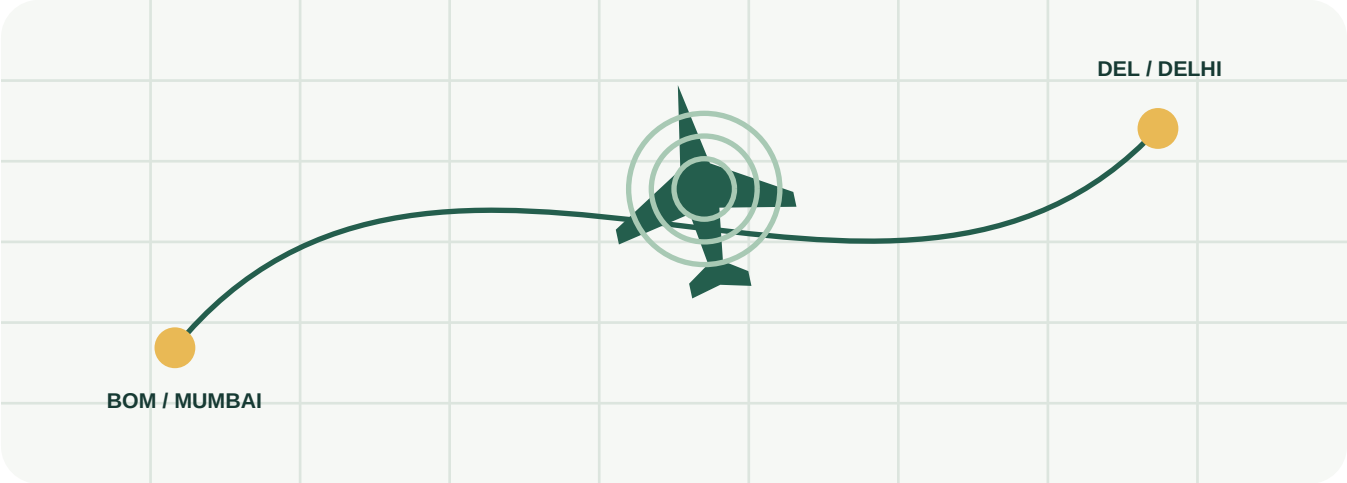
PROJECT & DELIVERY BLUEPRINT

Udaan

A flight search and tracking experience for journeys across India.



WHAT IT DOES	WHAT IT USES	WHAT COMES NEXT
Compare fares, airport boards and flight-number status.	Next.js, React, TypeScript, FlightAPI and Aviationstack.	Plan accounts, prepare hosting and deploy.

This PDF separates current features from future plans. Authentication is proposed here only and is not implemented in the app.

Prepared from the Udaan source tree and README. Deployment is planned, not completed.

01 / PRODUCT & FLOW

What Udaan does

Udaan combines flight shopping with flight status. Travellers compare itinerary choices, check airport arrivals/departures, or look up one flight number.

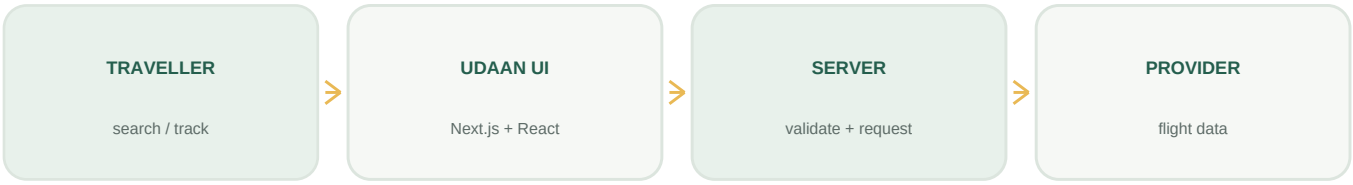
FEATURE	CURRENT BEHAVIOR	USER VALUE
Fare search	Search one-way/round-trip routes in India. Compare airlines, stops, duration and seller prices. Reuse saved results or request fresh FlightAPI data.	Compare, then follow a seller link to book.
Airport board	Choose an Indian airport and arrivals/departures. Narrow results by IST time and refresh manually.	Focused board for an airport and time range.
Flight lookup	Search Aviationstack by flight number; see status and details, with a map if coordinates exist.	Inspect one flight directly.
Booking	Fare options link to external sellers; Udaan does not take payment or issue tickets.	Discovery and comparison.

Technology

LAYER	TECH	ROLE
Web	Next.js 16, React 19, TypeScript	Pages, interface and server API routes.
UI	Tailwind CSS 4 + project CSS	Responsive styling.
Data	FlightAPI + Aviationstack	Fares and flight status/positions, subject to plan and coverage.

Positions may be missing or delayed. The current app does not continuously animate or poll aircraft locations.

System flow



Browser -> server validation -> provider request using server-side key -> normalized response in the UI. Keys are not sent to the browser.

02 / DATA & AUTHENTICATION

Providers and account proposal

DATA	SOURCE	USE / LIMIT
Fare offers	FlightAPI	Schedule/fare/seller data. Prices change; refresh may use credits.
Airport status	Aviationstack	Status/flight lookup. Coverage, quotas and endpoint access depend on plan.
Map position	Aviationstack	Map coordinates when returned; not guaranteed live tracking.
Future moving map	ADSB.lol - not integrated	Requires access plus engineering and ODbL 1.0 license review.

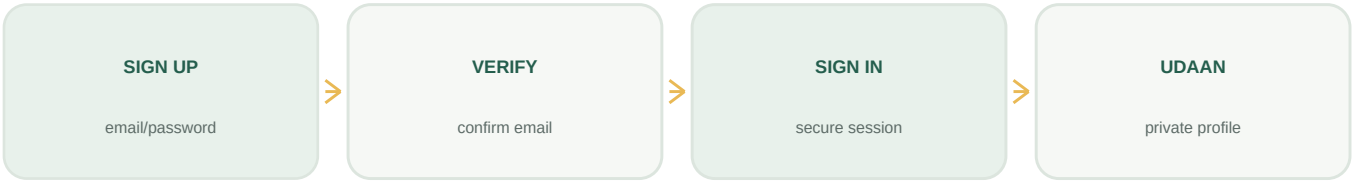
Security and privacy

Use .env.local locally and host environment settings in production. Exclude secrets and caches from Git. Restrict access to production keys, rotate exposed keys and avoid logging full provider URLs. Tell users their searches are sent to flight-data providers.

Authentication - proposal only

Signup/login are included only as a plan in this PDF. No authentication code, package or files have been added to Udaan.

Possible flow: create account -> verify email -> sign in -> access personal saved routes/preferences -> sign out.



BUILD	SAFE DESIGN
Auth service	Choose a maintained provider/library compatible with Next.js; do not hand-roll password storage.
Persistence	Use managed storage only if profiles/saved searches need persistence; minimize personal data.
Access	HTTP-only secure sessions; authorization checks in server routes.
Testing	Test verification, reset, logout, rate limits and protected endpoints.

03 / DEPLOYMENT

Roadmap to public release

Deployment has not been performed. A Next.js host such as Vercel is one option. Work through these gates before launch.

PHASE	TASK	GATE
Prepare	Review Git; include intended code/docs only. Exclude keys, cache and unrelated IDE files.	Clean branch; no secrets committed.
Verify	Install from lockfile; run npm run lint, npm test and npm run build; confirm provider plans.	Build passes; API access understood.
Connect	Import GitHub repository to Vercel or another Next.js host; check build settings.	Preview deployment.
Configure	Set FLIGHTAPI_KEY and AVIATIONSTACK_ACCESS_KEY for Preview/Production. Never prefix NEXT_PUBLIC_.	Server-only credentials configured.
Preview QA	Test mobile/desktop, search/refresh, airport boards, IST filters, flight lookup, errors and missing positions.	Preview checks pass.
Release	Attach domain, enable HTTPS, publish, review logs, quotas and monitoring.	Release owner assigned.

Production decisions

Local disk fare cache may not persist on serverless hosting; choose durable shared cache or accept temporary results. Confirm provider quotas before promising coverage. Keep refresh manual unless terms/budgets support polling. Add monitoring and a privacy notice.

Docs: Vercel Next.js deployment | environment variables | Aviationstack | ADSB.lol / ODbL

04 / RELEASE CHECKLIST

What is needed to ship

A public release needs valid provider access, a tested build and an operational owner.

READY	CHECK	WHY
[ ]	Confirm API keys active and plans allow each endpoint.	Prevent provider permission errors.
[ ]	Run lint, tests and production build.	Catch release problems.
[ ]	Configure secrets in hosting; redeploy after edits.	Keep credentials private and available server-side.
[ ]	Choose durable cache behavior for serverless.	Local disk may not persist.
[ ]	Test error/empty states and mobile layout.	Handle data gaps gracefully.
[ ]	Publish privacy, support and data-source info.	Set user expectations.
[ ]	Decide separately whether to implement auth.	Authentication is only proposed.
[ ]	Before ADSB.lol, verify access and ODbL terms.	Check attribution/share-alike requirements.

Describe Udaan as a flight search and status tool, not guaranteed real-time aircraft radar or a ticketing service.

Summary

Udaan compares domestic fares and provides airport status and flight-number lookup. FlightAPI supplies fares; Aviationstack supplies status and positions when available. Next: deployment readiness, durable caching, privacy disclosures and optional authentication. ADSB.lol moving-flight tracking is a future integration pending access and license review.

Project sources: README.md and .env.example. See linked provider/hosting documentation on the previous page.